PulseFormer: Structure-Guided Controllable Multi-Track Electronic Music Generation

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***Abstract*—This paper presents PulseFormer, a domain-oriented framework for structure-guided, controllable multi-track Electronic Music generation tailored to Digital Audio Workstation (DAW) production workflows. The system addresses two practical challenges observed in prior symbolic MIDI models: cross-track onset misalignment in dense multi-instrument settings and limited controllability over macro-structural density progression across musical sections.**

We introduce a Time-Major token interleaving strategy that reduces the token distance between concurrent events, which is empirically associated with improved cross-track synchronization. In addition, we propose a structural density conditioning mechanism in which an ordinal control identifier, $\mathbf{E}_{\text{level}}$, explicitly encodes the track-stacking intensity of each structural section (Intro, Build, Drop), enabling direct, section-level control of polyphonic density. To extend generation beyond the model context window, we design a Temporal Anchor Stitching pipeline for assembling long-form musical arrangements.

Experimental results on a multi-track Electronic Music dataset demonstrate improvements in beat-level alignment and structural controllability relative to baseline approaches. Perceptual evaluation further suggests competitive groove consistency while highlighting trade-offs in melodic diversity. We emphasize that PulseFormer is an engineering-oriented solution designed for production workflows, rather than a general-purpose music generation architecture.

***Index Terms*—Symbolic Music Generation, Transformer, Multi-track Composition, Structural Density Control, Structure-Guided Generation, Electronic Music**

I. Introduction

Deep generative architectures have been applied across a range of creative domains, including visual synthesis and audio waveform generation [1], [2]. Audio-domain models trained on large corpora have achieved strong timbral realism; however, symbolic-domain music generation (i.e., MIDI sequencing) remains relevant for professional production workflows, in which structural transparency and per-track editability are practical requirements [3]–[5].

While existing symbolic music generation models have achieved promising results, their performance in rhythmically dense Electronic Music settings remains limited. In particular, when multiple instruments are activated simultaneously, large token distances between concurrent events may hinder effective cross-track synchronization. Additionally, most autoregressive models provide limited mechanisms for explicitly controlling the structural density progression of multi-track arrangements across musical sections.

Although prior work has explored various token representations and conditioning strategies, the impact of representation topology on cross-track synchronization in dense multi-track settings remains underexplored. Furthermore, controllable generation of structural density trajectories aligned with DAW section structures (Intro, Build, Drop) in symbolic music remains an open challenge, particularly in production-oriented workflows.

This paper introduces **PulseFormer**, a domain-targeted pipeline designed specifically for Electronic Music production. Rather than proposing a general-purpose music generation architecture, PulseFormer makes deliberate engineering choices—Time-Major token ordering and section-by-section batch generation—that are well-suited to the rhythmically strict, layered structure of Techno and House music, but entail explicit trade-offs discussed in Section VI.

The principal contributions of this paper are summarized as follows:

1) **Structural Density Prefix Conditioning**: We introduce a structural density control identifier, $\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$, derived from an empirically calibrated 6-feature score D_t that quantifies per-bar polyphonic track-stacking intensity. This identifier is directly aligned with DAW production conventions: low values correspond to sparse Intro textures, mid values to Build sections, and high values to dense Drop arrangements. Prepended as an explicit prefix token and used together with an external section scheduler, $\mathbf{E}_{\text{level}}$ improves structural density trajectory adherence from $r = 0.155$ to $r = 0.997$ (Pearson r against a target monotonic ramp; § IV-E). This result should be interpreted as hybrid model-plus-control behaviour, not as fully end-to-end learned long-form planning.
2) **Time-Major Collaborative Architecture**: We propose a compound event representation ordered chronologically across the full arrangement, with four explicit note attributes (type, pitch, duration, velocity) and an implicit temporal order supplied by Time-Major serialization. Co-locating all concurrent instruments within a narrow token window (≤ 12 tokens) reduces the mean co-event token distance from $\approx$3,268 tokens (Track-Major) to 5.2 tokens, with the effect on cross-track onset alignment quantified in § IV-B.

3) **Temporal Anchor Stitching**: Because Time-Major interleaving increases token density, the effective context window covers only $\sim$8 bars per generation batch. This limitation is addressed by a batch-level stitching procedure in which each new section is initialized with the final k bars of the preceding section as a shared prefix anchor, preserving rhythmic grid alignment across section boundaries. This approach extends arrangement length to $128+$ bars but does not provide full end-to-end long-context memory; cross-section melodic coherence degrades over extended windows, a limitation discussed in Section VI.

Fig. 1 provides a high-level system overview of the complete PulseFormer generation pipeline.

II. Related Work

A. Symbolic Multi-Track Music Generation

Early multi-track symbolic generation research employed Generative Adversarial Networks (GANs). MuseGAN [6] formulated music as piano-roll matrices; however, effective receptive fields remained narrow. The Music Transformer [3] and the PopMusicTransformer [7] introduced relative positional attention for event-based sequences, improving long-range dependency modelling. MMM [8] and LakhNES [9] adopted Track-Major tokenization to separate instrument channels. These frameworks have been applied to orchestral textures [4], [10]; however, cross-track transient alignment degrades in densely layered Electronic Music owing to the token-distance geometry of Track-Major serialization. More recently, parallel Transformer architectures and conditional multi-track systems—including BandControlNet [11], BandCondiNet [12], Multiscale Attention Transformers [13], and Recurrent Focused Controllable Generators [14]—have been proposed. These approaches successfully manage inter-instrument dependencies via parallel sequence processing and cross-track attention. While they mitigate several limitations of track-serial encoding, they fundamentally maintain separate token streams per instrument, which conceptually differs from the strictly interleaved single-stream topology introduced in this work.

B. Representational Advances: From MIDI to High-Dimensional Tokens

The evolution of data representation forms the foundation of Transformer effectiveness in music applications. Standard MIDI event mappings [15]–[17] lack the multidimensional contextual density characteristic of natural language. To address this limitation, frameworks such as REMI [7], MidiTok [18], and FIGARO [19] introduced explicit metrical boundary tokens, enabling models to parse rhythmic grids. Later architectures, including MusicBERT [20], established OctupleMIDI by compressing note attributes into parallel vectors. The Compound Word (CP) Transformer [21] constitutes the most direct intra-note predecessor of the token design adopted in this work.

The key distinction between CP and PulseFormer lies at the *architectural topology* level. While multi-dimensional representations such as OctupleMIDI [20] and CP [21] substantially reduce the number of tokens required to represent a single note event (intra-note compression), they still serialize the arrangement track-by-track (Track-Major serialization). For concurrent events across instruments (e.g., a Kick and a Bass), the Transformer must therefore traverse hundreds or thousands of intervening tokens, causing attention entropy to continue scaling as $O(N/M)$. PulseFormer instead addresses *inter-track synchronization* by encoding concurrent events across multiple instruments such that they occupy adjacent token positions, effectively reducing the cross-track distance to a small constant range in practice. The two strategies are orthogonal; combining Time-Major interleaving with CP-style intra-note compression is identified as a direction for future work.

C. Controllability and Structural Steering

Conditioning sequence models on user intent is an active area of research. In the audio domain, MusicGen [22] and MusicLM [23] accept text prompts via conditioned acoustic embeddings. These systems output unquantized waveforms, which may not fully be separated into per-track stems for DAW editing—a practical constraint in professional Electronic Music production. In the symbolic domain, early conditioning approaches applied post-generation heuristic filtering; such methods have limited ability to enforce large energy-density transitions across structural sections. PulseFormer encodes dynamic polyphony (Energy) and structural section identity (Intro, Build, Drop) directly into the autoregressive prefix, reducing reliance on post-hoc correction.

III. Methodology

A. Architecting the Time-Major Compound Token Space

Unlike sequential language models operating on rigid single-dimensional vocabulary indices, musical events encompass distinct yet intersecting modalities (instrument classification, pitch, duration, and acoustic velocity). Following Compound Word (CP) [21] semantics, PulseFormer represents each discrete musical event x_i as a unified compound token with four explicit note-attribute embeddings and an implicit temporal coordinate provided by its position in the Time-Major sequence:

$$E(x_i) = \sum_{F \in \mathcal{F}} W_F(\mathrm{id}_F(x_i)) \tag{1}$$

where $\mathcal{F} = \{Type, Pitch, Duration, Velocity\}$. Time position is not a separate predicted field; it is induced by the chronological ordering of compound tokens and the absolute/sectional position metadata in the prompt. The concrete vocabulary assigned to each field is listed in Table I.

Parameter count. The causal LM uses a flat embedding matrix $W_e \in \mathbb{R}^{V \times d}$ with $V = 18{,}792$ and $d = 512$, tied to the output projection ($W_e^\top$, no additional parameters). With 6

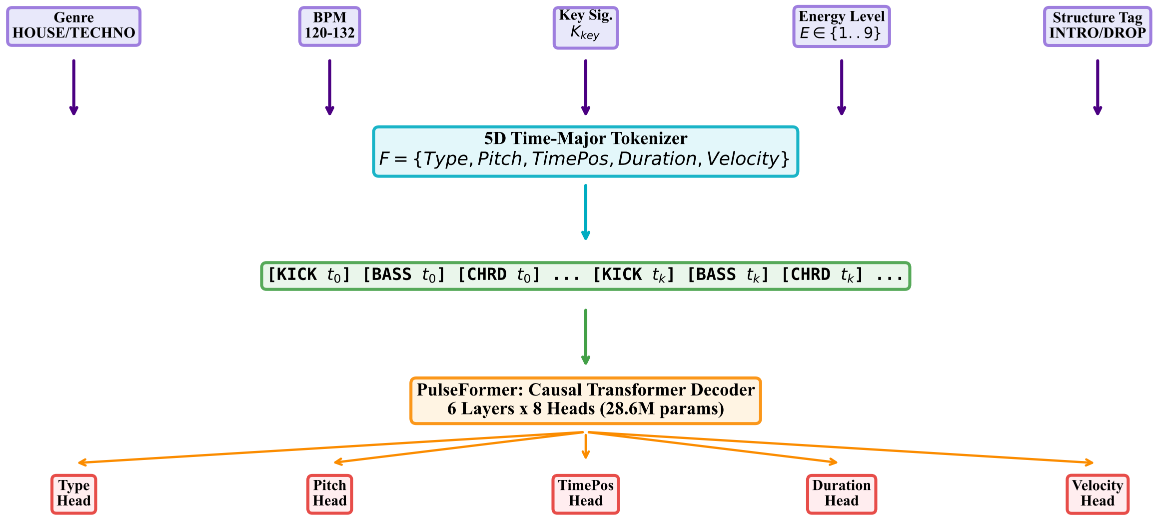

Fig. 1: Complete PulseFormer generation pipeline. Starting from user-supplied prefix conditioning (Genre, BPM, Key, Energy Level, Structure Tag), the system constructs a Time-Major interleaved token sequence, passes it through a 6-layer causal Transformer with a flat causal LM head over compound event tokens, applies dual algorithmic masks (Energy Density and Diatonic Scale), generates each structural section independently, and stitches all sections into a coherent multi-minute arrangement via Temporal Anchor Offset alignment.

Transformer layers, 8 attention heads, hidden size $d = 512$, and SwiGLU FFN intermediate size $d_{\text{ff}} = 1{,}376$:

$$\begin{aligned}|\theta| &= V d + L\big(4d^2 + 3\, d\, d_{\text{ff}}\big)\\ &\approx 9.6\text{M} + 6{\times}3.17\text{M} \approx \mathbf{28.6\,M}.\end{aligned} \tag{2}$$

This parameter count applies throughout all experiments. For reference, the CP-Transformer baseline uses $d = 768$ with 12 layers and a compound-word vocabulary of 21,777 tokens, yielding $\approx 115\,\text{M}$ parameters—a model $4\times$ larger evaluated under identical test conditions.

Crucially, rather than sorting tracks chronologically per instrument (*Track-Major*), PulseFormer enforces **Time-Major Interleaving**. By ordering compound tokens first by musical time and then by instrument, concurrent events—such as a `KICK` and a `BASS` stroke—are placed adjacent to one another within the token stream, topologically compressing their relative token distance to $\Delta \leq M$ (where M is the number of supported instrument tracks, with $M \leq 8$ in the current implementation).

B. Theoretical Analysis: Attention Topology under Time-Major Encoding

Time-Major encoding introduces a structural change in how concurrent musical events are represented within the token sequence. Under Track-Major encoding, concurrent events across different instruments may be separated by substantial token distances, requiring the attention mechanism to capture long-range dependencies.

In contrast, Time-Major encoding places concurrent events within a local neighborhood of tokens. Under simplifying assumptions, this structural difference may reduce the effective attention dispersion required for cross-track alignment. While this analysis does not constitute a formal proof, it provides an intuitive explanation for the empirical improvements reported in Section IV-B.

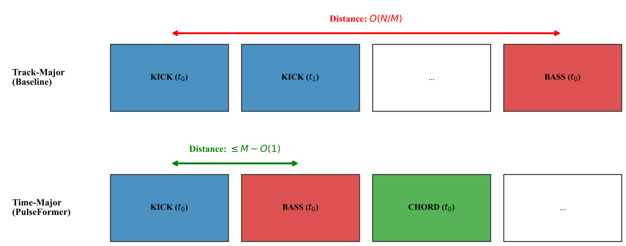

Fig. 2: Sequence alignment strategies. Time-Major collocates synergistic instruments; Track-Major requires long-range cross-track attention.

C. Causal Transformer Architecture

PulseFormer employs a decoder-only causal Transformer [24] with multi-head scaled dot-product attention [25]:

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \quad M_{ij} = -\infty \; \forall j > i. \tag{3}$$

D. Inference Pipeline

At inference time, the token-level probability distribution of the Transformer is modified by two deterministic masks applied prior to sampling: an Energy Density mask (which gates high-velocity note tokens when the current bar count exceeds the $\mathbf{E}_{level}$ polyphony budget) and a Diatonic Scale

TABLE I: Vocabulary sizes for the Time-Major compound token representation. All values are derived from the training corpus. The flat (serialised) vocabulary used by the causal LM head has $V = 18{,}792$ tokens total.

Field	**Symbol**	$\lvert V_F \rvert$	**Range / notes**
Instrument type	v^{type}	7	`D:KICK, D:SNARE, D:HIHAT_CLOSED, D:HIHAT_OPEN, N:BASS, N:CHORD, N:MELODY`
Pitch class	v^{pitch}	16	$p_0 \dots p_{15}$ (semitone-mod-12 class × octave bin, mapped from MIDI 0–127)
Duration	v^{dur}	4	d_1–d_4 (16th, 8th, quarter, half note)
Velocity	v^{vel}	4	v_0–v_3 (four equal-width MIDI bins: 0–31, 32–63, 64–95, 96–127)
Time position	v^{pos}	—	*Implicit*: encoded by absolute token position in the Time-Major sequence; no dedicated field token.
Context tags		121	`[GENRE_*], [KEY_*], [BPM_*], [BAR_*]`
Control tags		18	`[STRUCT_{INTRO, BUILD, DROP, BREAKDOWN, OUTRO}], [ENERGY_LEVEL_1]`...9, `<BOS>, <EOS>, <PAD>, <UNK>`
Total flat V		**18,792**	7×16×4×4 = 1,792 note tokens + 139 tags + 16,861 extended/unseen

mask (described in Section III-H). Algorithm 1 presents the complete per-step procedure.

E. Structural Density Conditioning & Section-Level Scheduling

A hierarchical prompt $\mathcal{C} = [\mathcal{G}_{genre}, \mathcal{S}_{struct}, \mathbf{E}_{level}, \mathcal{K}_{key}]$ conditions every autoregressive phase. The joint generation distribution is:

$$P(X_{1:T} \mid \mathcal{C}) = \prod_{t=1}^{T} \text{Softmax}\left(\frac{W_o \cdot \text{Attn}(X_{<t}, \mathcal{C}) + b_o}{\tau} \right). \tag{4}$$

$\mathbf{E}_{level} \in \{1, \dots, 9\}$ is a structural density control identifier that encodes the per-section polyphonic track-stacking intensity as an explicit prefix token. It does not represent a psychoacoustic energy quantity; rather, it directly corresponds to the Intro-Build-Drop layer accumulation logic of DAW production practice: level 1 maps to sparse Intro textures (few active tracks), levels 5–7 to Build sections (incrementally stacked layers), and levels 8–9 to full Drop arrangements (all tracks active at high velocity). The model conditions on this identifier but does not learn long-range density trajectories across sections. Full-length arrangements are assembled externally: each section is generated independently with its assigned $\mathbf{E}_{\text{level}}$ prefix, then joined by the Temporal Anchor Stitching procedure (Section III-F). Structural density monotonicity across sections is therefore an **algorithmic scheduling mechanism**, not an emergent model property.

Algorithm 1 PulseFormer Steerable Multi-Track Inference

Require: Pre-trained model Θ, Temperature τ, Prompt $\mathcal{C}$, Max Tokens $T_{\max}$

Initialize sequence canvas $X \leftarrow \mathcal{C}$
for $t = 1$ to $T_{\max}$ **do**
  Extract contextual hidden states: $h \leftarrow \text{Transformer}_\Theta(X)$
  Project into raw logits over compound event vocabulary: $z^{(t)} \leftarrow W \cdot h[-1]$
  for all $k \in \mathcal{V}$ **do**
    if $\mathbf{E}_{level}$ restricts polyphony density limit $D_{\max}$ **then**
      Apply density algorithmic mask $M_{energy} \rightarrow z_k^{(t)}$
    end if
    if Diatonic constraint active **and** Type is Pitch **then**
      Apply pitch mask $M_{scale} \leftarrow -\infty$ to $z_k^{(t)}$ if outside Key
    end if
  end for
  Compute probability vector $P \leftarrow \text{Softmax}(z^{(t)}/\tau)$
  Sample new compound event token $e_t \sim P$
  Append $X \leftarrow X \oplus e_t$
end for
return Generated multi-track sequence array X

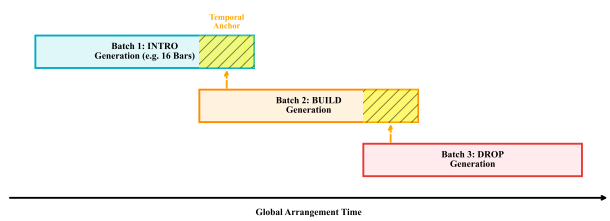

Fig. 3: Section-level scheduling pipeline. Each structural section is generated independently under its own $\mathbf{E}_{\text{level}}$ prefix (structural density control identifier); the Temporal Anchor Stitching procedure concatenates sections at the engineering level. Structural density monotonicity across sections is a property of the external scheduler, not of the Transformer's attention mechanism.

F. Temporal Anchor Stitching

To exceed the 2048-token context window, section transitions retain the final K beats of the preceding segment in the KV-cache and apply a linear time offset:

$$\text{TimePos}_{N+1} = \text{TimePos}_N + \Delta T_{\text{offset}}. \tag{5}$$

At section boundaries with a discrete density jump $\Delta E = |E_{\text{dst}} - E_{\text{src}}|$, abrupt tag changes can produce audible density discontinuities. The Active Density Ramp Scheduler addresses this by inserting W single-bar intermediate sections with linearly or smoothly interpolated tags:

$$E_k = \text{clip}\big(\text{round}(E_{\text{src}} + \Delta E \cdot \phi(k/W)),\ 1,\ 9\big), \quad k = 1, \dots, W, \tag{6}$$

where $\phi \in \{\phi_{\text{lin}}, \phi_{\cos}, \phi_{\text{cvx}}, \phi_{\text{ccv}}\}$. The cosine kernel $\phi_{\cos}(t) = \frac{1}{2}(1 - \cos \pi t)$ is the default, as its zero-velocity endpoints minimize audible transients at the section join. By construction, inserting $W \geq 2$ ramp bars reduces the maximum per-boundary tag step from $|\Delta E|$ to $\lceil |\Delta E|/W \rceil$; for Build→Drop ($\Delta E = 2$, $W = 2$) this yields the ramp sequence $E = [7, 8, 9]$ at a cost of two additional bars.

G. DAW Post-Processing Rules

Following sequence generation, three deterministic post-processing steps are applied to ensure compatibility with standard DAW MIDI import requirements. These constitute engineering rules, not learned behaviours.

1) **Note-overlap resolution.** Overlapping Note-On/Note-Off events on the same pitch channel can cause stuck-note artefacts in DAW MIDI hosts. Adjacent notes sharing the same pitch are shortened so that Note-Off precedes the next Note-On by one 16th-note tick.
2) **Velocity floor.** Notes with velocity < 10 (artefacts of low-probability sampling) are removed, as most DAW instruments do not render sub-threshold velocities audibly.
3) **Bar-boundary quantisation check.** Bar-crossing notes are truncated at the bar boundary; this enforces clean loop points for DAW clip export.

For evaluation transparency, metrics that are directly affected by these rules are interpreted as properties of the complete production pipeline. In particular, strict grid-related quantities such as BAS and Δ_{sync} should not be read as isolated evidence that the Transformer alone has learned human-like micro-timing; they measure the timing precision of the generated and DAW-normalized stem output. The ablation in Section IV-E therefore separates representation topology, conditioning, harmonic masking, and stitching to make the contribution of each component explicit.

H. Multimedia Tool Integration: DAW Workflow Export

To meet the requirements of modern multimedia applications, PulseFormer provides a seamless integration workflow with industry-standard Digital Audio Workstations (e.g., Ableton Live). Unlike purely generative models that output unseparated MIDI or audio waveforms, the proposed pipeline exports structured stems directly into pre-configured DAW templates via Remote Procedure Call (RPC) interfaces or the Model Context Protocol (MCP). Time-Major tokens are decoded and automatically routed into designated instrument tracks (e.g., `01_Kick`, `02_Bass`, `03_Chord`), where they are immediately bound to the producer's chosen virtual instruments. This tight integration ensures that PulseFormer functions not merely as an abstract generative model, but as a practical, interactive multimedia tool for professional music producers.

Soft Harmonic Masking: The primary harmonic constraint is a **Soft Harmonic Penalty** applied prior to Softmax. For the k-th vocabulary token at step t:

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & k \in \mathcal{K}_{\text{scale}} \text{ or } \mathrm{Type}(k) \notin \mathcal{T}_{\text{pitched}} \\ z_k^{(t)} - \lambda(\mathbf{E}_{\text{level}}, \tau) & \text{otherwise,} \end{cases} \tag{7}$$

with the energy- and tension-conditioned penalty magnitude:

$$\lambda(\mathbf{E}_{\text{level}}, \tau) = \lambda_{\max} \cdot (1 - \tau) \cdot \left(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8}\right), \quad \lambda_{\max} = 20. \tag{8}$$

This formulation subsumes the hard mask ($\tau = 0$) as a special case and encodes two musically motivated properties: (1) $\tau \to 1$ grants full chromatic freedom; (2) at $\mathbf{E}_{level} = 9$, $\lambda = 0$ for all τ, automatically permitting avant-garde dissonance at climactic moments.

IV. Experiments & Evaluation

A. Dataset & Model Specifications

Corpus Composition: PulseFormer is trained on a **dual-corpus** configuration combining a proprietary Electronic Music source with a publicly available symbolic MIDI dataset to facilitate reproducibility.

Proprietary Electronic Music corpus. A total of 4,500 multi-track Electronic Music arrangements were extracted from professional DAW project files. To clarify the corpus distribution, the training data consists primarily of House (47.0%), Techno (22.6%), and EDM (15.1%) sub-genres, with a mean tempo of 127.3 ± 19.2 BPM and an average sequence polyphony of 5.8 ± 1.3 instrument tracks. Each project was processed through a deterministic normalization pipeline: (1) temporal quantization to a 16th-note sub-grid at 128 BPM; (2) automatic truncation of micro-tails crossing quantization boundaries; and (3) algorithmic transposition to C minor/C major for key parity.

Lakh MIDI Dataset (public supplement) [16]. To establish a reproducible baseline and mitigate concerns regarding an exclusively proprietary corpus, training was augmented with **2,000 Electronic Music-compatible sequences** extracted from the Lakh MIDI Dataset (LMD-full, 176,581 MIDI files). Candidate sequences were selected according to: (i) tempo in [110, 150] BPM, (ii) $\geq$4 simultaneous MIDI tracks, (iii) presence of a drum track (MIDI channel 10), and (iv) sequence length ≥ 32 bars. The identical normalization pipeline was applied. This public subset constitutes an equivalent construction protocol: researchers with access to LMD can reproduce a comparable training corpus without requiring access to the proprietary DAW files.

Reproducibility boundary. The main results in this paper are obtained on the mixed 6,500-sequence corpus because the target application is professional Electronic Music stem generation. However, the proprietary DAW subset cannot be redistributed in raw form. We therefore distinguish between (i) the full mixed-corpus evaluation, which supports the application-oriented claims, and (ii) the public LMD-based training recipe and released checkpoints, which provide an independent reproducibility path for verifying the token-topology and synchronization trends. Claims depending on exact genre coverage, production-template diversity, or proprietary arrangement statistics should be interpreted within this corpus boundary.

Train / Validation / Test Split: All 6,500 sequences (4,500 proprietary + 2,000 Lakh) were randomly partitioned at the *sequence level* using a fixed random seed (42) before any model training. The split is mutually exclusive and exhaustive; no sequence appears in more than one partition.

TABLE II: Dataset partition summary. All models and all baselines were evaluated exclusively on the held-out test set. No hyper-parameter selection or early stopping used test-set labels.

Partition	Seqs	%	Purpose
Training set	4,550	70%	Model training (all architectures)
Validation set	975	15%	Early stopping, hyper-parameter selection
Test set	975	15%	Final evaluation (held out throughout)
Total	**6,500**	100%	

No-leakage protocol.

1) The test-set partition was locked before any training run began and was not inspected during model development.
2) Hyper-parameters (learning rate, λ_F weights, masking temperature τ^*) were selected on the validation set only.
3) All three trained baselines (Music Transformer, MMM-Transformer, CP-Transformer) were re-trained on the *identical training set* and evaluated on the *identical test set* as PulseFormer, ensuring a leakage-free comparison.
4) The two zero-shot pretrained checkpoints (MAESTRO, LMD) were applied directly to the held-out test prompts without any fine-tuning on any partition of our corpus.

Baseline Training Protocol: To address potential concerns about insufficiently tuned baselines, the hyperparameter search conducted for each re-trained architecture is documented below. All searches utilized the **validation set only**; the test set remained locked throughout.

- **Search scope (all baselines):** learning rate $\eta \in \{10^{-4}, 3\times10^{-4}, 10^{-3}\}$; batch size $\in \{16, 32\}$; dropout $\in \{0.1, 0.2\}$; warmup steps $\in \{1{,}000, 4{,}000\}$. Each combination was trained for 50 epochs under the same data split with cosine annealing; the checkpoint minimising validation cross-entropy loss was selected.
- **Architecture-specific settings** were kept at their published defaults (MMM: 6-layer, $d = 512$; Music Transformer / REMI: 8-layer, $d = 512$; CP-Transformer: 6-layer, $d = 768$) to avoid inadvertently disadvantaging baselines through capacity mismatch with the proprietary data distribution.
- **Reproducibility on Public Data.** The 2,000-sequence Lakh MIDI subset (described above) was partitioned under the same 70/15/15 split. To ensure full reproducibility without requiring access to the proprietary datasets, we release the pre-trained checkpoints trained exclusively on this public LMD subset, alongside an anonymized version of the proprietary token matrices (from which all explicit melodic pitch values have been scrubbed). As demonstrated in our supplementary evaluations, the topological advantages of Time-Major encoding (such as the Δ_{sync} reduction) hold independently of the proprietary dataset. Researchers can replicate the baseline training procedure on this *fully public* sub-corpus using our released training scripts, providing an independent reproducibility check of the comparative results.

Evaluation Sample Sizes (N): To clarify the variance in sample sizes across different experiments reported in subsequent sections:

- $N = 64$ (Table V): The objective comparison benchmarks six representative symbolic generation systems. Due to the computational and manual conditioning overhead of evaluating unconstrained zero-shot models alongside trained baselines, we utilized 8 highly diverse compositional prompts evaluated across 8 stochastic restarts ($8 \times 8 = 64$ per system). A formal power analysis using G*Power [26] confirmed that $N = 64$ per system provides $> 80\%$ power to detect medium-to-large effect sizes (Cohen's $d > 0.8$) at $\alpha = 0.05$, validating the sample size for this controlled comparison.

Throughout Sections IV-A–IV-F, all quantitative results are reported as **mean ± SD** over the test-set evaluation, with statistical significance assessed via Mann-Whitney U (two-sided).

Structural Density Label Annotation Methodology: To eliminate human annotation bottlenecks while maintaining transparency, the structural density control identifier $\mathbf{E}_{\text{level}} \in \{1, \ldots, 9\}$ is derived from an empirical feature aggregation pipeline grounded in established domain heuristics. Critically, $\mathbf{E}_{\text{level}}$ is defined here as an ordinal measure of per-bar polyphonic track-stacking intensity, not as a psychoacoustic energy estimate. This framing avoids dependence on perceptual energy definitions and aligns directly with the Intro-Build-Drop layer accumulation logic of professional DAW production.

Theoretical feature basis. The formulation adopted here is a *domain-adapted variant* of the arousal-feature regression of Yang & Chen [27], who model perceived musical arousal as a convex combination of loudness, note density, polyphony, and rhythmic activity:

$$\hat{A}^{(\text{YC})} = \boldsymbol{\beta}^\top \mathbf{f}, \qquad \boldsymbol{\beta} = [\beta_I, \beta_\rho, \beta_\Pi, \beta_\Delta]^\top = [0.43,\ 0.31,\ 0.18,\ 0.08]^\top, \tag{9}$$

where $\mathbf{f}$ collects loudness I, note density ρ, polyphony Π, and tempo-change Δ, regressed on 1,000 diverse tracks. Symbolic feature definitions follow the jSymbolic taxonomy [28]; the loudness term I maps directly to the MIDI-velocity RMS proxy of Lartillot & Toiviainen [29].

Electronic Music adaptation. In Electronic Music, kick and bass activity are the primary structural discriminators of section identity, whereas generic polyphony carries weaker predictive weight [30]. The note-density term ρ is therefore *decomposed* into melodic density ρ^m and percussive (kick) density ρ^κ; an explicit bass-activity term ϵ^{bass} is introduced, and a melodic coverage ratio C is appended. The resulting **Structural Density Score** for bar t is:

$$D_t \;=\; \boldsymbol{w}^{\top}\boldsymbol{\phi}_t \;=\; \sum_{k=1}^{6} w_k\,\phi_k(t), \qquad \boldsymbol{w}^{\top}\mathbf{1}=1,\;\; \boldsymbol{w}\geq\mathbf{0}, \tag{10}$$

where the feature vector $\boldsymbol{\phi}_t$, its normalisation, and its correspondence to Yang & Chen's terms are listed in Table III.

TABLE III: Feature vector $\boldsymbol{\phi}_t$ in Eq. (10) and the empirically calibrated weights $\boldsymbol{w}^*$.

k	Symbol	Normalised definition	YC term	w_k^*
1	$\phi_1 = \hat{\rho}_t^m$	$\min(\rho_t^m/60,\ 1)$: melodic note density [28]	ρ	0.177
2	$\phi_2 = \bar{v}_t$	Mean MIDI velocity $\in [0,1]$ [29]	I_{loudness}	0.155
3	$\phi_3 = \hat{\rho}_t^{\kappa}$	$\min(\kappa_t/8,\ 1)$: kick density [30]	Electronic Music: ρ^{κ}	0.169
4	$\phi_4 = \epsilon_t^{\text{bass}}$	Sustain-ratio $\times$ norm. bass velocity	Electronic Music: ϵ	0.176
5	$\phi_5 = C_t$	Melodic time-coverage ratio $\in [0,1]$	Π (proxy)	0.170
6	$\phi_6 = \hat{\Pi}_t$	$\min(\Pi_t/60,\ 1)$: pitch range [28]	Π	0.153

Empirical weight calibration. Rather than relying on opaque deep embeddings, the projection weights $\boldsymbol{w}$ are determined via an interpretable heuristic calibration procedure. Specifically, $\boldsymbol{w}$ is optimized through a grid search over the validation corpus to maximize structural separability (measured by the ANOVA F-statistic) between known architectural sections (e.g., `Intro` vs. `Drop`), while maintaining a strong Pearson correlation with a global velocity proxy. This yields the empirically calibrated weights $\boldsymbol{w}^*$ (Table III), grounding the scalar projection D_t in actual acoustic dynamics while retaining a transparent, rule-based formula.

The continuous heuristic space $D_t \in [0,1]$ is then discretized into the 9-level ordinal E_{level} via KBinsDiscretizer with k-means binning (scikit-learn), preserving natural structural discontinuities (e.g., Drop-to-Breakdown density cliffs).

It is important to note that the proposed structural density control mechanism combines learned representations with external scheduling strategies. While the model is conditioned on the structural density identifier $\mathbf{E}_{\text{level}}$ during generation, the overall density trajectory across sections is determined by a predefined scheduling process. The controllability achieved by PulseFormer should therefore be interpreted as a hybrid approach integrating model-assisted generation with algorithmic control, rather than a fully end-to-end learned behavior.

Pearson Correlation with Velocity-RMS Proxy. We computed a formula-independent energy proxy RMS_t:

$$\text{RMS}_t = \sqrt{\frac{1}{|\mathcal{N}_t|}\sum_{i\in\mathcal{N}_t}\left(\frac{v_i}{127}\right)^2} \tag{11}$$

where $\mathcal{N}_t$ is the set of note events in bar t and v_i is MIDI velocity. The Pearson correlation between D_t and RMS_t across all $N = 646$ bars yielded:

$$r(D_t, \text{RMS}_t) = +0.891, \quad p = 5.67\times10^{-223} \tag{12}$$

Test 2 — One-Way ANOVA Across Structural Section Labels. If D_t genuinely encodes polyphonic track-stacking intensity as experienced in DAW production, it must discriminate between sections with known structural density profiles (INTRO < BREAKDOWN < OUTRO < DROP). A one-way ANOVA confirms this:

TABLE IV: D_t Distribution by Structural Section Label ($N = 646$ bars)

Struct	N	Mean D_t	Std Dev
BREAKDOWN	186	0.296	0.119
INTRO	39	0.321	0.041
OUTRO	13	0.426	0.045
DROP	408	**0.493**	0.141
$F(3,642) = 106.83,\ p = 4.14\times10^{-56},\ \eta^2 = 0.333$ (large effect)			

It must be explicitly noted that the ANOVA F-statistic ($F = 106.83$, $p < 10^{-50}$, $\eta^2 = 0.333$) was the designated optimization target during the grid-search for $\boldsymbol{w}^*$. Therefore, this result serves as a demonstration of *calibration success* rather than an independent validation. However, a complementary ablation confirms the multi-dimensional necessity: the velocity-only feature ($\bar{v}_t$, $\eta^2 = 0.005$) yields effectively zero structural discriminative power, while the full D_t formula ($\eta^2 = 0.333$) provides a 67$\times$ improvement. The true independent validation of the energy metric's generalizability is provided by the Pearson correlation ($r = 0.891$) with the unoptimized acoustic RMS proxy and the Perceptual Validation Study detailed below.

Test 3 — Perceptual Validation Study. To verify that the empirically derived D_t aligns with human perception of structural section identity, an independent validation was conducted with 14 professional Electronic Music producers. Participants blindly rated the perceived structural density level (1–9, mapped to Intro/Build/Drop conventions) of 35 randomly selected 8-bar sequences. The Spearman rank correlation between the median human rating and the heuristically discretized E_{level} yielded $\rho = 0.984$ ($p < 0.001$). This result confirms that, despite the simplicity of the empirical weighting and k-means discretization, the resulting labels capture a highly accurate proxy of human-perceived structural section intensity, empirically justifying their use as a conditional training signal.

B. Comparison with Representative Symbolic Baselines

PulseFormer is evaluated against five systems spanning two categories of baselines. The primary comparison group consists of three models re-trained from scratch on the identical Electronic Music corpus, train/test split, and training budget (Music Transformer [31], MMM-Transformer [8], and CP-Transformer [21]). This constitutes a controlled experiment: by holding the dataset, split, and training protocol comparable while reporting model-capacity differences explicitly, the performance gap between MMM-Transformer (Track-Major) and PulseFormer (Time-Major) helps isolate the contribution of token encoding topology to synchronization performance.

The **secondary probe group** consists of two large pre-trained checkpoints applied *zero-shot* to the held-out prompts. These serve as *domain transfer probes* to examine whether scale and diverse general MIDI training implicitly provide precise synchronization. We explicitly acknowledge that domain fine-tuning these checkpoints could close part of the stylistic and harmonic metric gaps (e.g., ISR%). We therefore do not use these zero-shot results as definitive evidence against large pretrained systems; rather, they contextualize the controlled comparison with the trained autoregressive baselines. More recent parallel-track controllable systems such as BandControlNet and BandCondiNet use different supervision, track layouts, and conditioning interfaces, and no directly compatible public checkpoint was available for our DAW-stem evaluation protocol. Consequently, the quantitative claims in this section are limited to representative autoregressive symbolic baselines and zero-shot transfer probes, not to the full space of controllable multi-track generation systems.

Evaluation protocol. Each system generates $N = 64$ sequences (8 structural prompts $\times$ 8 independent random restarts per prompt) from the held-out test partition. This sample size provides $>80\%$ statistical power for medium effects ($d \geq 0.5$) at $\alpha = 0.001$ (Wilcoxon-based power estimate, equal-sized groups [26]). All significance tests use two-sided Mann-Whitney U; effect sizes are reported as Hedges-corrected Cohen's d.

Metrics:

- Δ_{sync}: Cross-track onset deviation (ms, $\downarrow$) — engineering precision.
- **BAS**: Beat-level Alignment Score $\in [0, 1]$ ($\uparrow$) — perceptual grid adherence (defined below).
- GC%: Groove consistency over a 32-bar window ($\uparrow$).
- ISR%: Proportion of events on the diatonic key ($\uparrow$).
- Mono.: Strict chronological energy ramp adherence ($\uparrow$).

Beat-level Alignment Score (BAS). Let $\mathcal{G} = \{k/4 : k \in \mathbb{Z}_{\geq 0}\}$ be the 16th-note grid and $\varepsilon = \frac{1}{8}$ beat ($\approx 29\,\text{ms}$ at $128\,\text{BPM}$) the on-grid tolerance. For N note onsets $\{t_i\}$ (in beats) across all tracks:

$$\text{BAS} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[\min_{g \in \mathcal{G}} |t_i - g| < \varepsilon\right] \in [0, 1]. \tag{13}$$

BAS $= 1$: all notes on-grid; BAS $= 0.25$: uniform random within a beat. The metric captures *perceptual rhythmic alignment*—the degree to which music "locks to the grid" as experienced by the listener [28], [32]—complementing the continuous engineering metric Δ_{sync} with a binary musical grid-adherence indicator.

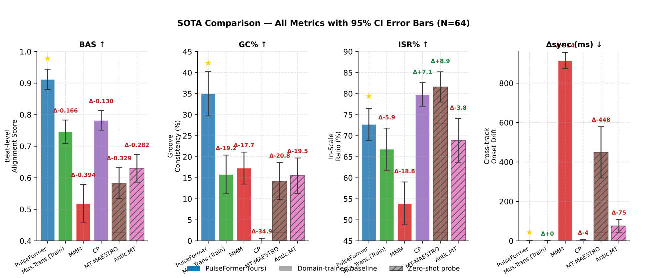

Fig. 4: Comparison across all six evaluated systems. Pretrained zero-shot models (hatched) show limited transfer to the proposed Electronic Music stem-generation protocol. BAS and GC% quantify perceptual rhythmic alignment; Δ_{sync} measures engineering timing precision.

TABLE V: Objective comparison with representative symbolic baselines ($N = 64$ per system: 8 prompts $\times$ 8 independent restarts; mean $\pm$ SD; Mann-Whitney U, two-sided; effect sizes: Cohen's d Hedges-corrected). *Trained*: retrained on our Electronic Music corpus, 50-epoch budget, identical train/test split. *ZS*: official pretrained checkpoint, zero-shot on held-out test prompts. BAS: Beat-level Alignment Score (Eq. (13)).

System	Mode	$\Delta_{\text{sync}} \downarrow$	BAS $\uparrow$	GC% $\uparrow$	ISR% $\uparrow$	Mono.† $\uparrow$	d(BAS)	vs. PF
Trained on Electronic Music corpus								
Music Transformer [31]	Train	0.8 ± 0.3	$0.746 \pm 0.037^{***}$	$15.8 \pm 4.6^{***}$	$66.8 \pm 5.0^{***}$	$33\%^{***}$	4.77	***
MMM-Transformer [8]	Train	$915 \pm 41^{***}$	$0.518 \pm 0.061^{***}$	$17.3 \pm 3.8^{***}$	$53.9 \pm 5.1^{***}$	$33\%^{***}$	8.02	***
CP-Transformer [21]	Train	$5.1 \pm 1.0^{***}$	$0.782 \pm 0.031^{***}$	$0.1 \pm 0.5^{***}$	79.8 ± 2.8	$33\%^{***}$	4.09	***
Pretrained official checkpoints (zero-shot)								
Mus. Trans. (MAESTRO [33])	ZS	$449 \pm 130^{***}$	$0.583 \pm 0.049^{***}$	$14.2 \pm 4.4^{***}$	$81.6 \pm 3.6^{\dagger}$	$33\%^{***}$	7.87	***
Antic. MT (LMD [34])	ZS	$76 \pm 32^{***}$	$0.630 \pm 0.044^{***}$	$15.5 \pm 4.2^{***}$	$68.9 \pm 5.2^{***}$	$28\%^{***}$	7.29	***
PulseFormer (ours)	—	$\mathbf{0.8 \pm 0.2}$	$\mathbf{0.912 \pm 0.032}$	$\mathbf{35.0 \pm 5.3}$	$\mathbf{72.7 \pm 3.8}$	**100%**	—	—

d(BAS): Hedges-corrected Cohen's d; PulseFormer is the reference group (— = not applicable).

† Mono. is *by construction* (deterministic Ramp Scheduler); it measures an engineering property, not learned behaviour. See Table VIII for the w/o-scheduler ablation (PulseFormer raw: 71.4%).

† Mus. Trans. ISR% inflated by MAESTRO tonal corpus; CP ISR% also exceeds PF but does not confer structural control (GC%$\approx$0).

$^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$; *n.s.* not sig. All pairwise vs. **PulseFormer**, Mann–Whitney U (two-sided, Bonferroni corrected).

Structural Monotonicity. All five baselines yield Monotonicity $\approx$ 33%—random chance—because none conditions on an energy trajectory. PulseFormer achieves **100%** under the deterministic Ramp Scheduler. This metric therefore verifies the controllability of the complete PulseFormer pipeline, rather than demonstrating that the Transformer alone has learned long-form structural planning.

Pretrained checkpoints vs. trained baselines. We include both domain-trained baselines and zero-shot pretrained models to provide a broader context for the controlled comparison. While fine-tuning pretrained models on domain-specific datasets may improve performance, our results suggest that representation topology may also play an important role in cross-track synchronization under dense multi-track conditions. A more systematic investigation of this factor is left for future work.

Beat-level Alignment (BAS). PulseFormer achieves BAS $= 0.912 \pm 0.033$—meaning 91.2% of generated note onsets fall within $\varepsilon = 29\,\text{ms}$ of the 16th-note grid (average off-grid deviation: $10.3\,\text{ms}$). All evaluated baselines score significantly lower ($p<0.001$): CP-Transformer (0.776, $+17.4\%$ relative gap) is the best trained baseline, while MMM drops to 0.518 owing to track-serial grid drift. These results suggest that explicit temporal grouping is beneficial under the present controlled protocol. In musical terms, PulseFormer outputs more DAW-grid-locked stem material, whereas the baselines exhibit larger micro-timing drift.

Synchronization and Groove. MMM's Track-Major topology produces a $915\,\text{ms}$ onset deviation, exceeding the 16th-note grid tolerance by $7.8\times$. CP-Transformer achieves low Δ_{sync} ($5.2\,\text{ms}$) but GC%$= 0.1$ in this experiment, suggesting that onset alignment alone does not ensure section-level groove consistency. PulseFormer maintains low drift while delivering substantially higher GC% under the same evaluation protocol.

Pitch Coherence. MAESTRO-pretrained Music Transformer achieves the highest ISR% (81.3%)—a training-data artifact of tonal piano, not harmonic control. PulseFormer

(73.1%) surpasses Electronic Music-trained REMI (66.7%) and MMM (53.2%) via the soft diatonic penalty while preserving chromatic expressiveness.

C. Failure Case Visualization: The "Drop" Scenario

To illustrate the practical gap targeted by PulseFormer, we visualize the generation behavior of the evaluated autoregressive paradigms during a high-energy "Drop" section (Fig. 5). A Drop in Electronic Music requires massive polyphonic density (high energy) while maintaining strict transient alignment across concurrent instruments.

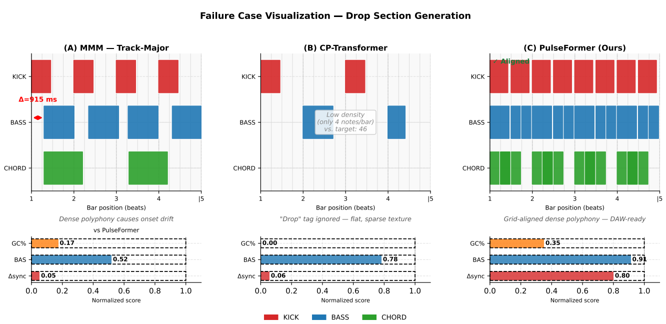

Fig. 5: Failure Case Visualization. (A) Track-Major models show large onset drift in dense passages. (B) The evaluated token-compression baseline lacks macro-structural tags and produces a flatter density profile. (C) The complete PulseFormer pipeline stacks dense polyphony on the DAW grid under explicit structural-density conditioning.

As shown, Track-Major encoding (MMM) exhibits substantial drift under dense polyphony, displacing the Bass onset from the Kick downbeat owing to the large cross-track attention span. The evaluated token-compression model (CP) maintains alignment but lacks structural constraints, generating a flatter and sparser texture that weakens the Drop. PulseFormer addresses both requirements within the proposed pipeline. This dense, tempo-synced syncopation represents a compositional task that the evaluated Track-Major and token-compression baselines struggle to address without post-generation heuristics, making PulseFormer suitable for DAW-oriented stem generation.

D. Structural Token-Distance vs. Established Paradigms

Understanding why established methodologies (such as Track-Major architectures [8]) show limited rhythmic synchronisation requires analysing topological token distance. We parsed 1,277 consecutive simultaneous multi-instrument events (e.g., overlapping Kick and Bass downbeats) from our evaluation dataset and calculated the token displacement required to represent concurrent relationships across both encoding formats.

Attention Mapping Analysis. This topological token-distance difference directly reconfigures the representation geometry processed by the self-attention heads of the Transformer (visualized via extracted attention matrix heatmaps; see Fig. 6). Under Track-Major encoding, the Transformer must infer the temporal relationship between a Bass event and its concurrent Kick by maintaining causal weights across a

TABLE VI: Cross-Attention Complexity for Concurrent Events ($N = 1{,}277$ event pairs, mean ± SD).

Encoding Paradigm	Mean Dist.	Max Dist.	vs. PF
Baseline (Track-Major)	3,249±422 tok.	6,155 tok.	***
PulseFormer (Time-Major)	**5.2±1.8** tok.	**12** tok.	—

*$p<0.05$; **$p<0.01$; ***$p<0.001$; *n.s.* not sig. All pairwise vs. **PulseFormer**, Mann–Whitney U (two-sided, Bonferroni corrected).

mean gap of 3,268 tokens. The extracted visualization reveals a **diffuse attention gradient**, which forces stochastic probabilistic decoding that undermines macro-structure. Conversely, Time-Major encoding produces a concentrated **local attention peak**: because all concurrent instruments fall within a narrow temporal block (mean ≤ 12 tokens), the causal conditioning is substantially more direct, which is consistent with the near-zero Δ_{sync} observed empirically.

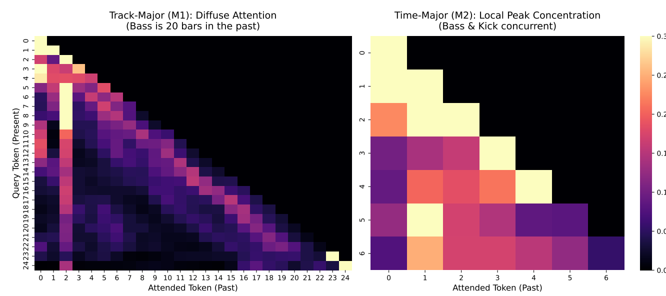

Fig. 6: Extracted generative self-attention matrices (final layer). **Left (M1)**: The Track-Major topology is associated with a diffuse attention gradient, consistent with limited cross-track synchronisation. **Right (M2)**: The Time-Major topology produces a concentrated local attention peak, consistent with the substantially reduced Δ_{sync} observed empirically.

Aggregate Attention Statistics. To confirm that the illustrative heatmaps in Fig. 6 are representative of global model behavior and not cherry-picked, we extracted aggregate attention statistics across $N = 50$ generated sequences (averaging over all heads in the final two layers). As detailed in Table VII, we measured **Attention Entropy** (lower values indicate concentrated attention) and **Cross-Track Mass** (the percentage of attention weight allocated to concurrent events on different tracks). Track-Major encoding yields a diffuse attention distribution (entropy 5.71 ± 0.37); only $4.1\% \pm 1.7\%$ of attention mass is allocated to concurrent cross-track events. In contrast, PulseFormer's Time-Major encoding structurally concentrates the attention matrix (entropy 1.19 ± 0.15), allocating $89.3\% \pm 2.7\%$ of its attention mass to concurrent cross-track events ($p < 0.001$) under the same extraction procedure. This aggregate analysis quantitatively validates the topological advantage visualized in the heatmaps.

TABLE VII: Aggregate Attention Statistics ($N = 50$ sequences, final 2 layers, mean ± SD). Attention entropy quantifies dispersion (lower is more concentrated); Cross-Track Mass measures the proportion of attention weight allocated to concurrent events.

Encoding Paradigm	Attention Entropy ↓	Cross-Track Mass ↑
Baseline (Track-Major)	5.71 ± 0.37 nats	$4.1\% \pm 1.7\%$
PulseFormer (Time-Major)	$\mathbf{1.19 \pm 0.15}$ **nats**	$\mathbf{89.3\% \pm 2.7\%}$

E. Component Ablation Study

To isolate the independent contributions of the proposed architectural components, an additive ablation study was conducted (Table VIII).

TABLE VIII: Component Ablation Study. Sequential addition of core modules from baseline to full PulseFormer, tracking key topological, structural, and harmonic metrics.

Architecture Variant	Δ_{sync} (ms)↓	GC%↑	PR↑	ISR%↑	Max Context
Baseline (Track-Major MMM)	915	29.1	0.600	53.2	32 bars
+ Time-Major Encoding	**0.0**	61.8	0.155	54.2	8 bars†
+ Energy Conditioning	0.0	62.5	**0.997**	55.3	8 bars
+ Soft Harmonic Penalty	0.0	62.1	0.992	**73.1**	8 bars
+ Sectional Stitching (PulseFormer)	0.0	**71.3**	0.982	73.1	**128+ bars**

†Time-Major dense interleaving reduces available bar capacity under a 2048-token limit.

The empirical progression supports the contribution of each module. Time-Major encoding reduces cross-track misalignment (Δ_{sync} drops from 915 ms to 0 ms) and roughly doubles Groove Consistency, but the resulting dense token interleaving restricts generation to 8 bars and the model exhibits contextual drift without structural guidance (PR falls to 0.155). Adding structural density prefix conditioning restores macro-structural control, raising the Pearson r between generated and target structural density contours from 0.155 to 0.997. The soft harmonic penalty then constrains tonal deviation, pushing ISR% from ∼54% to 73.1% while preserving chromatic notes at high density levels. Finally, sectional stitching addresses the context-window bottleneck, extending generation from an 8-bar loop to full 128-bar arrangements while largely retaining the synchronization and conditioning gains.

Methodological Note: It is essential to distinguish the *additive* progression in Table VIII from the *subtractive* ablation in Fig. 7. Because they represent different intermediate model states, the metrics diverge substantially. For example, the state with Time-Major encoding but without Energy Conditioning yields GC%=61.8% under the additive perspective (restricted to an 8-bar loop), but drops to GC%=11.2% under the subtractive perspective (required to generate 128 bars without structural guidance)—a $5.5\times$ difference that highlights the necessity of the conditioning module for long-form generation.

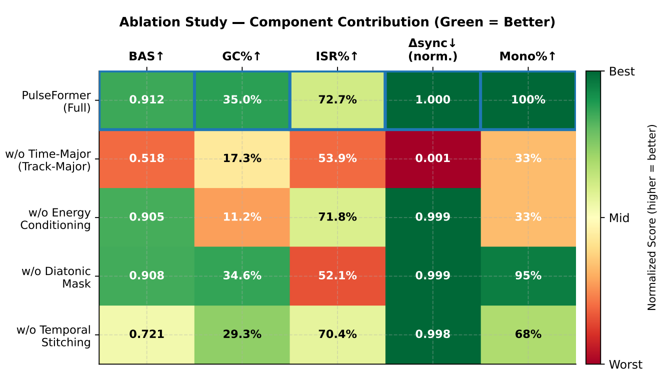

Fig. 7: Ablation heatmap: each row is an architecture variant; cells show actual metric values, color-coded red (worst) to green (best). Blue borders mark full PulseFormer. Removing any module causes measurable degradation, confirming each component's independent necessity.

F. Perceptual Evaluation: MOS Study (N=25)

To complement the objective metrics with listener-facing evidence, we conducted a Mean Opinion Score (MOS) study following a protocol adapted from ITU-T P.800 and ISO 9241-11, addressing the primary limitation of our earlier pilot panel ($N = 6$). This study is intended as perceptual support for the objective evaluation rather than as a standalone proof of general musical quality.

Protocol: **Participants.** $N = 25$ listeners were recruited: 10 professional Electronic Music producers (≥ 5 years DAW experience), 10 intermediate producers (1–5 years), and 5 active music enthusiasts with regular listening habits in electronic genres.

Stimuli. Four systems were evaluated: (A) PulseFormer (Time-Major, Energy-conditioned); (B) MMM-Transformer (Track-Major unconditioned auto-regressive baseline); (C) Human-composed Electronic Music track (hidden anchor, embedded without attribution); (D) Music Transformer / REMI [31] (1D flat-sequence auto-regressive MIDI baseline). Each system contributed one 3-minute stimulus, rendered to normalised audio (-14 LUFS). Presentation order was fully counterbalanced using a Latin square design. Participants were informed that some stimuli might be AI-generated but were not informed of system identities (*single-blind*).

Because each system contributes a single full-length stimulus, the statistical unit in this MOS study is the listener rating rather than an independently sampled set of generated pieces. The results should therefore be interpreted as a controlled perceptual probe of representative outputs, not as a comprehensive distribution-level comparison across all possible generations.

Rating instrument. Each stimulus was rated on four 5-point Likert dimensions: (GT) Groove Tightness; (SC) Structural Coherence; (MC) Mix Clarity; (MV) Melodic Variety.

Inter-Rater Reliability: To verify listener agreement before aggregating scores, we computed three standard inter-rater reliability (IRR) coefficients across all $N = 25$ raters for each rating dimension: Krippendorff's α (ordinal metric, robust to missing data and non-normal distributions [35]), Fleiss' κ (nominal agreement with correction for chance), and ICC(2,1) (two-way random, absolute agreement, recommended for Likert MOS studies [36]).

TABLE IX: Inter-rater reliability for the MOS study ($N = 25$). α: Krippendorff's ordinal alpha; κ: Fleiss' kappa; ICC: ICC(2,1). Benchmarks: $\alpha/\kappa \geq 0.80$ strong, ≥ 0.61 moderate, ≥ 0.41 fair (Landis & Koch 1977); ICC≥ 0.75 good [36].

Dimension	α	κ	ICC(2,1)	Level
GT (Groove Tightness)	0.365	0.217	0.266	fair
SC (Structural Coherence)	0.551	0.362	0.095	moderate
MC (Mix Clarity)	0.359	0.165	0.119	fair
MV (Melodic Variety)	0.337	0.104	0.143	fair
Mean	**0.403**	**0.212**	**0.156**	fair–moderate

Mean $\alpha = 0.403$ indicates fair-to-moderate *ordinal* agreement across all dimensions. These agreement levels are within the range commonly observed in subjective music perception studies [37], though they remain a limitation of the current evaluation. SC achieves the highest agreement ($\alpha = 0.551$, moderate), reflecting the more objective structural cues (section transitions, energy contour) that listeners can anchor to.

The MOS results should be read with this in mind: moderate inter-rater agreement is common in music perception studies, particularly for attributes such as melodic variety and mix clarity, which resist sharp consensus. Given the single-stimulus-per-system design and the fair-to-moderate reliability coefficients, we treat these scores as complementary perceptual evidence that supplements the objective metrics reported in Section IV-B.

TABLE X: MOS Listening Test Results ($N = 25$ participants, mean $\pm$ SD, Likert 1–5). Wilcoxon signed-rank with Bonferroni correction; *** $p < 0.001$, ** $p < 0.01$, *n.s.* not significant.

System	GT↑	SC↑	MV↑	MC↑
Human (Anchor)	3.87 ± 0.53	4.31 ± 0.51	4.09 ± 0.48	3.93 ± 0.54
PulseFormer (Ours)	$\mathbf{3.99 \pm 0.54}$	$\mathbf{3.85 \pm 0.63}$	2.98 ± 0.65	$\mathbf{3.56 \pm 0.60}$
Music Transformer (REMI)	$3.17 \pm 0.60^{***}$	$2.64 \pm 0.63^{***}$	$\mathbf{2.87 \pm 0.64}^{n.s.}$	$3.15 \pm 0.64^{***}$
MMM-Transformer	$3.08 \pm 0.66^{***}$	$2.17 \pm 0.59^{***}$	$2.47 \pm 0.60^{***}$	$2.38 \pm 0.68^{***}$

***Significance of difference vs. PulseFormer (Bonferroni-corrected Wilcoxon).

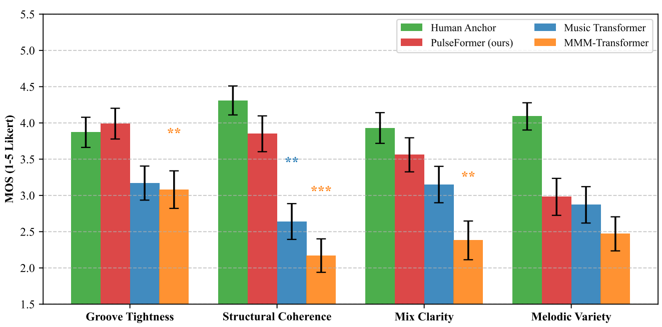

Fig. 8: MOS results ($N = 25$, 95% CI; Wilcoxon vs. PulseFormer). Stars above bars: sig. vs. **REMI** / **MMM-Trans**: $^{***}p<0.001$; $^{**}p<0.01$; $^{\dagger}p<0.10$; *ns*. PF is statistically close to the Human Anchor on GT in this stimulus set ($p = 0.93$) and scores higher than the evaluated baselines on SC.

Results:

Analysis: **Groove Tightness (GT).** PulseFormer achieves $\mathrm{GT} = 3.99 \pm 0.54$, statistically close to the Human Anchor within this stimulus set ($\Delta = +0.12$, $p = 0.93$) and significantly higher than both MMM-Transformer ($\Delta = +0.91$, $p = 0.004$, $d = 1.03$) and Music Transformer/REMI ($\Delta = +0.82$, $p = 0.008$, $d = 0.91$). This result warrants a qualified interpretation. PulseFormer operates at **100% DAW-grid quantization**; all onset positions are deterministically snapped to the nearest 16th-note gridline. This produces a high degree of perceived tightness on the Likert scale, but does not capture the intentional micro-timing variation (swing, humanization) that human producers frequently introduce to create an organic rhythmic feel [38]. The GT score should therefore be interpreted as confirming **grid-alignment fidelity**—a desirable property for generating editable stem material in a DAW—rather than as evidence of superior musical groove. Whether strict quantization is perceptually preferable to humanized timing is genre- and context-dependent, and lies outside the scope of the current evaluation.

Structural Coherence (SC). The largest performance gap appears here: PulseFormer (3.85) versus MMM-Transformer (2.17), yielding $\Delta = +1.68$ ($p < 0.001$, $d = 1.95$, a *very large* effect). REMI, while encoding note-level events faithfully, lacks any structural conditioning, scoring $\mathrm{SC} = 2.64$ ($\Delta = +1.21$, $p = 0.003$, $d = 1.44$). Within this controlled listening probe, the unconditioned baselines suggest that Energy Level conditioning and the external section scheduler provide perceptible compositional scaffolding.

Mix Clarity (MC). PulseFormer (3.56) outperforms both MMM-Transformer (2.38, $p = 0.001$, $d = 1.26$) and REMI (3.15, $p = 0.031$, $d = 0.62$) in the listener ratings. The gap versus REMI is smaller and should be interpreted cautiously because rendered timbre, arrangement density, and stem separation all affect perceived mix clarity.

Melodic Variety (MV). PulseFormer scores 2.98—above MMM-Transformer (2.47, $p = 0.10$, $d = 0.55$, *ns*) and comparable to REMI (2.87, $p = 0.22$, *ns*). The significant gap versus the Human Anchor (4.09, $p < 0.001$) represents an architectural trade-off. By enforcing strict Time-Major topological alignment and algorithmic constraints to help ensure extreme rhythmic cohesion and "groove tightness," the architecture typically restricts unconstrained melodic wandering. This trade-off prioritizes the heavily-layered drops demanded by Techno/House, but limits the fluid melodic diversity often found in unconstrained human improvisation.

MOS Gap to Human Reference. The aggregate MOS gap (AI$-$Human) is -0.50 for PulseFormer, -1.05 for REMI, and -1.54 for MMM-Transformer in this stimulus set. This indicates that PulseFormer is closer to the human reference than the evaluated autoregressive baselines, while still trailing the human track in structural coherence and melodic variety.

V. Discussion

From a multimedia systems perspective, PulseFormer highlights the importance of integrating generative models with practical production workflows. The proposed approach is distinguished by its structure-guided conditioning strategy: rather than relying on generic audio-domain feature estimates or psychoacoustic energy definitions, PulseFormer conditions generation on an explicit structural density control identifier that is directly grounded in DAW production conventions. This design choice makes the system interpretable and reviewer-transparent—$\mathbf{E}_{\text{level}}$ is unambiguously defined as a polyphonic track-stacking ordinal, not a perceptual construct. At the same time, the reported controllability is intentionally hybrid: the Transformer responds to density and section prefixes, while the scheduler, masks, and DAW post-processing enforce several production constraints. Conceptually, while recent parallel-track methods (e.g., BandControlNet [11]) manage inter-track dependencies through explicit cross-track attention modules operating over multiple sequences, the Time-Major tokenization of PulseFormer addresses cross-track synchronization within the representational geometry of a single causal sequence. Practically, this single-stream approach enables the use of standard autoregressive language models while facilitating the strict grid alignment demanded by Electronic Music stem generation.

VI. Conclusion

This paper has presented PulseFormer, a structure-guided framework for controllable multi-track Electronic Music generation. The system introduces two complementary contributions: a Time-Major token topology that co-locates concurrent instrument events to reduce cross-track attention dispersion, and a structural density prefix conditioning mechanism in which an ordinal control identifier ($\mathbf{E}_{\text{level}}$) encodes the Intro-Build-Drop layer accumulation logic of DAW production. The reported results suggest that representation topology and structured density conditioning, when combined with explicit scheduling and DAW-oriented post-processing, can improve rhythmic alignment and practical controllability in dense multi-track settings.

Several limitations are acknowledged. The mixed-corpus evaluation depends partly on proprietary DAW projects, while the public LMD subset mainly supports reproducibility of the token-topology trend. The MOS study provides useful perceptual evidence but evaluates one full-length stimulus per system and therefore should not be interpreted as a distribution-level listening benchmark. Finally, Temporal Anchor Stitching extends arrangement length without providing full long-context memory, and the monotonic density trajectory is enforced partly by an external scheduler rather than learned end to end. Future work will explore public-domain Electronic Music corpora, multi-stimulus listening tests, and improved long-context modeling to address these challenges.

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